

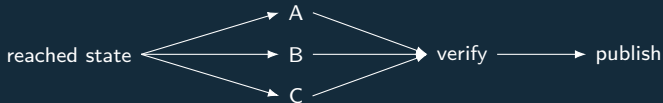
Forkable Sandboxes

**Forkable sandboxes reuse execution state.
Each result needs its own verification.**

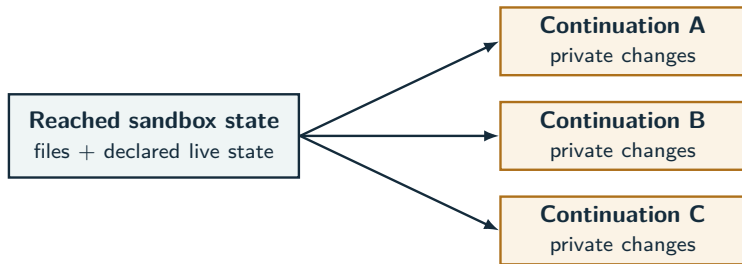
Yossi Eliaz

Principal Engineer, Incredibuild · Associate Professor, Holon Institute of Technology (HIT)

Disclosure: I work for a company that builds sandboxes.



A sandbox fork starts private continuations from one reached state.



- Preserve the state the next action uses.
- Keep each child's writes private.
- Choose worktrees, warm files, or a process/VM checkpoint as required.

Fork contract; examples: SnowFlock, Kimi AgentENV, and DSec.

Additional requests reduced latency, and additional samples raised coverage.

BigTable request hedging

1,000-key read; backup after 10 ms.

1,800 → **74 ms**

p99.9 latency

2% additional requests

SWE-bench Lite sampling

DeepSeek-Coder-V2-Instruct

Samples	Coverage
---------	----------

1	15.9%
---	-------

250	56%
-----	------------

Coverage still requires selecting a correct candidate.

Dean & Barroso, CACM 2013; Brown et al., arXiv:2407.21787.

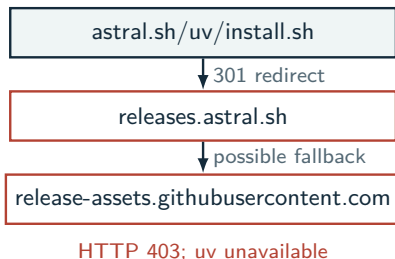
We recorded two factory runs and implemented evidence checks.

Factory record	Elapsed time	Model cost
Allowlist update and dispatch repair	43:44	\$10.33
Separate delivery/recovery incident	~26 min	\$3.43

- Repeated evidence IDs stop the numerical merge.
- Lineage travels with each numerical result.
- The factory gate binds approval to an artifact digest.

SELFHOST-2/3 records; Eliaz, arXiv:2607.09689.

The installer fix added two redirect hosts across a six-file plan.



- **islo failure:** six allowed hosts, two missing destinations.
- **Factory:** Airflow + Docker Sandboxes; hosted model through Databricks.
- **Plan:** six files; matching declarations and hermetic regression checks.

SELFHOST-2 intent, plan, and stage records; 27 September 2026.

Repairs and reviews consumed 59% of the recorded factory stage time.

Stage	Work performed	Time	Cost
Specification and plan	Acceptance conditions; six-file scope	3:41	\$0.96
Build and test	Three edits; suite failure; first repair	20:01	\$4.50
Review	Review; second repair; tests; re-review	18:55	\$4.87

43:44 elapsed 42:38 recorded stages **\$10.33 model cost**

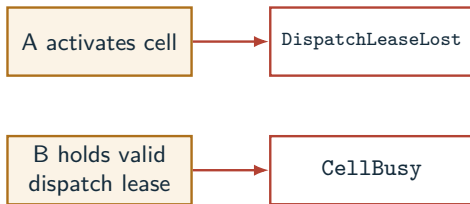
Two repairs + two reviews: 25.2 min, \$7.75; 59% of stage time, 75% of cost.

Setup: **20.3 s = 0.79%** of stage time. Calculated 100× speedup: **20.1 s saved**.

SELFHOST-2 metrics and agent-session records; reported costs at list price.

The repair added same-owner adoption without its required direct test.

Recorded diagnosis: 0 runs from 4 replicas



Required test: force same-owner
CellBusy; assert exactly one dispatch.

- **Fix:** after same-work-order CellBusy, adopt the active cell.
- **1,972 reported cases**, including skips; **0 failures**.
- Review approved without the direct test; `service.py` changed outside the six-file plan.

SELFHOST-2 first repair, second repair, review, and edit-hook records.

Delivery refused three times, then cleanup erased an approved patch.



Separate work order: **about 26 min, \$3.43, no delivered PR.**

Recorded object	Actual limitation
-----------------	-------------------

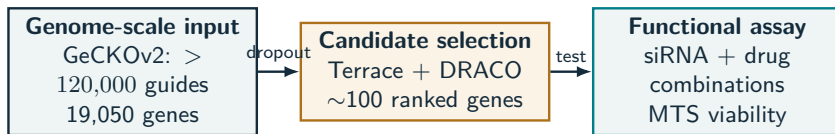
CI success	Evaluation skipped without a key; job passed in about 3 s.
Human approval	Harness answered; record named a human actor.
Adapter digest	Shim source stayed outside Git and was not recoverable.

DESIGN Export the patch, input identity, and typed receipts before teardown.

SELFHOST-3 delivery records; SELFHOST-2 adapter/approval records; CI, 28 September 2026.

Six of eight drug combinations improved killing in MCF-7 cells.

Goal: find gene perturbations that increase sensitivity to the SRC-3 inhibitor SI-12.



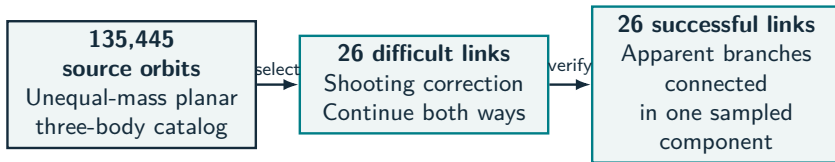
Replication unit: each viability point had ≥ 4 technical replicates; each plot represented ≥ 2 independent experiments.

Rank candidates, then measure effects in a separate assay.

Gilad, Eliaz et al., *Communications Biology* 4:399 (2021), Figs. 2–4.

26 bidirectional continuations joined the projected orbit branches.

With Ori Chamo: test whether projected branches of periodic three-body orbits connect in solution space.



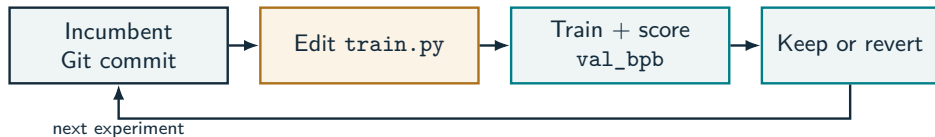
Tools: Extra-Trees proposes warm starts; shooting restores periodicity; Floquet calculations assess stability.

Independent 60-digit checks confirmed representative stability transitions.

Chamo–Eliaz, [ai.viXra:2608.0069](https://arxiv.org/abs/2608.0069) (2026); Three-Body Orbit Atlas records. Preliminary preprint.

Autoresearch kept 23 of 126 experiments and lowered its score by 2.8%.

Karpathy's autoresearch: edit `train.py`, train for five minutes, then compare validation bits per byte.



126 attempts: 23 kept, 102 discarded, 1 crash.

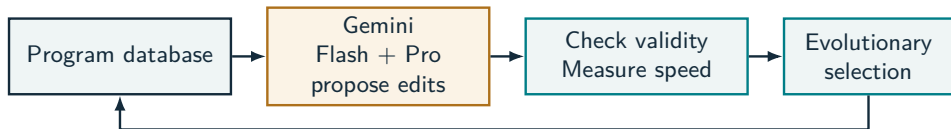
Best score: 0.997900 $\rightarrow$ 0.969686; lower is better.

A fixed evaluator turns code generation into a hill-climbing loop.

autoresearch protocol; maintainer's reported session. Five minutes is the training budget per completed attempt.

AlphaEvolve's scheduler recovered 0.7% of Google's worldwide compute.

Gemini proposes programs; automated evaluators determine which programs enter the population.



Borg scheduling heuristic

0.7% average worldwide compute recovered.

Gemini training kernel

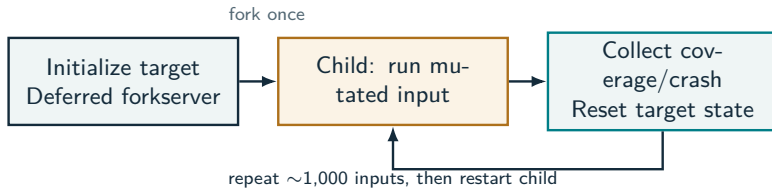
23% kernel speedup → 1% shorter training.

Keep both the local gain and the end-to-end gain in the result record.

Google DeepMind, AlphaEvolve (2025); reported production applications.

AFL++ reports 10–20 $\times$ speedups from persistent process reuse.

Coverage-guided fuzzing mutates inputs, runs the target, and retains inputs that reach new behavior.

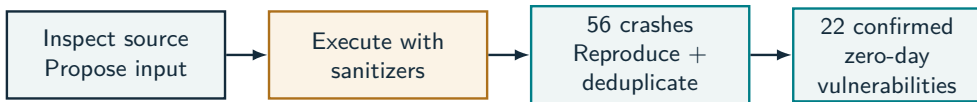


Forking saves initialization; resetting correctly makes persistent reuse useful.

AFL++ [persistent-mode documentation](#): typical 10–20 $\times$ speedup; about 1,000 loop iterations recommended.

CyberGym's GPT-5 campaign turned 56 crashes into 22 confirmed zero-days.

OpenHands + GPT-5 explored current code from 431 OSS-Fuzz projects: 1,748 executables.



Historical benchmark: 1,507 vulnerabilities in 188 projects; the input crashes the vulnerable version and does not crash the patched version.

An executed crash is a lead; validation establishes the security finding.

Berkeley RDI, CyberGym; CyberGym report. Latest-code campaign and historical benchmark are separate experiments.

Worktrees, checkpoints, and replay preserve different kinds of state.

State to reuse	Tools	What a continuation receives
Source/build state	Git worktrees, OBuilder	Files and cached build inputs
Process state	CRIU, gVisor save/restore	Supported process/kernel state
VM state	Xen, SnowFlock, Firecracker	Declared memory and device state
Recorded observations	rr, DSec replay	Recorded execution or responses

Choose the mechanism for the state the continuation actually consumes.

Applying a code patch may require reload or restart.

CRIU, gVisor, Firecracker and rr documentation; SnowFlock, EuroSys 2009; DeepSeek-V4 §5.2.5.

DeltaBox checkpoints in 10.83 ms; SnowFlock forks remotely in 600–800 ms.

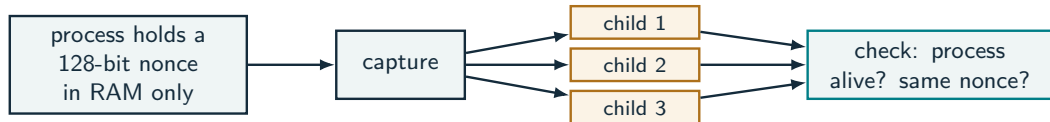
System	Operation reported	Value	Source
DeltaBox	checkpoint	10.83 ms	arXiv 2026
Kimi K3 AgentENV	checkpoint	as low as 133 ms	tech. report §5.3.2
Kimi K3 AgentENV	resume	as low as 49 ms	tech. report §5.3.2
Shepherd	fork	134–143 ms	arXiv 2026
SnowFlock	remote fork across hosts	600–800 ms	EuroSys'09
Firecracker	boot to application code	under 125 ms	NSDI'20

Time the operation that delivers the state your continuation needs.

DeltaBox; Kimi K3 §5.3.2; Shepherd; Lagar-Cavilla et al., EuroSys'09; Agache et al., Firecracker, NSDI'20.

A RAM-only 128-bit nonce tests memory restoration in three children.

Experiment design: Gate 0, followed by declared semantic checks.



Result

Permitted description

files survive only

filesystem restore

process and RAM survive

evidence of process and memory restoration

declared semantic checks match

fidelity for those checked properties only

A child passes the memory probe by retaining the live process and nonce.

Our preregistered protocol, Gate 0. [Kimi K3 §5.3.2](#). CRIU and Firecracker snapshot documentation.

Each child needs private writable state and host-controlled authority.

Inherited or shared	Required contract
Immutable parent/cache	Digest identity; declared trust domain
Writable branch state	Private to the child
Child permissions	Host authenticates and authorizes identity, epoch, expiry and budget
Publishing credential	Controller holds it; approval names the artifact

Kernel isolation does not make a copied bearer token harmless.

Firecracker and gVisor documentation; Capsicum, USENIX Security 2010; CHERI, IEEE S&P 2015.

Restoring local state does not undo completed external actions.

Effect	What restore cannot undo
Sent model request	The remote invocation and its cost
Sent tool request	A remote action that may have completed
Staged patch	Local bytes can be retained; publication needs a separate gate

- Proposed effect journal and buffer: reconcile uncertainty or use idempotency.
- Export artifacts and receipts before every teardown.

Speculator, SOSP 2005; external synchrony, OSDI 2006; Remus, NSDI 2008; DeepSeek-V4 §5.2.5.

Forking saves resources when avoided preparation exceeds reuse overhead.

$$(N - 1) P > H + N (R + D)$$

preparation P , capture H , restore R , divergence D , children N . Additive resource-cost model.

Illustrative resource cost in resource-seconds.

Case	Avoided preparation	Added reuse cost	Result
$N=8, P=30, H=2, R=1, D=3$	$7 \times 30 = 210$	$2 + 8 \times 4 = 34$	fork saves 176
same, warm cache $P=2$	$7 \times 2 = 14$	34	fork loses 20

Batch elapsed time is a separate question: it depends on critical paths and contention. A faster-booting minimal image is the alternative (Jitsu, LightVM).

Madhavapeddy et al. NSDI'15 (Jitsu); Manco et al. SOSP'17 (LightVM, NEC Labs). Illustrative parameter values.

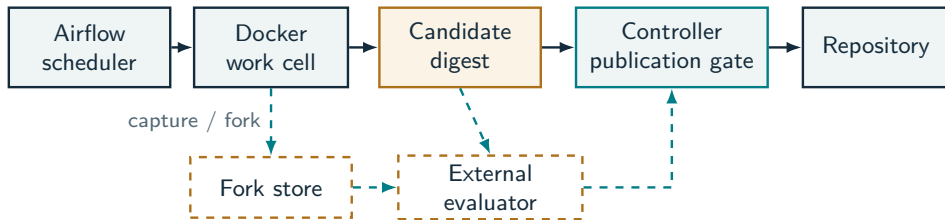
islo completed 255/256 restore–run–capture calls, median 6.87 s.

Workflow	Success	p50	p95
Daytona: create	1,024/1,024	0.20 s	1.12 s
Tensorlake: create	256/256	3.44 s	9.00 s
islo: restore, run, capture	255/256	6.87 s	9.04 s

- Restore path: a 141 MB named snapshot; concurrency 12.
- Creation paths: requested concurrency 8.
- Each row names its timing endpoint; teardown is excluded.

Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689), Table 2.

The controller binds repository publication to a candidate digest.

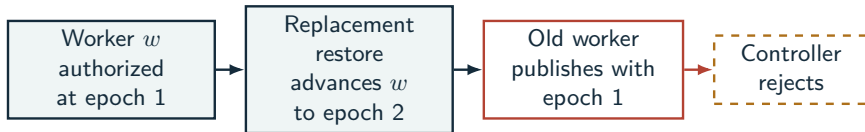


Solid: serial components. Dashed: fork and evaluation with artifact-bound receipts.

Returned candidates and receipts meet at one controller-held gate.

Factory records and design; arXiv:2607.09689v4.

An epoch check rejects the old instance after worker replacement.



- Approval binds one frozen artifact.
- Concurrent fork gets a new worker identity and its own epoch.

The controller checks authority outside the restored guest state.

Proposed promotion protocol; Gray & Cheriton, Leases, SOSP 1989.

Selection, composition, and pooling require different evidence.

Returned objects	Decision	Required evidence
Different patches	Select one	Validate the selected artifact
Compatible patches	Compose	Test the exact composition
Measurements of one target	Reduce	Calibrated information and dependence model

The implemented numerical reducer occupies the measurement lane.

A patch score is not automatically a precision estimate.

Factory records; Evidence-Aware Reduction for Forkable Compute, arXiv:2607.09689v4.

Checkpoint-started success is conditional on reaching that checkpoint.

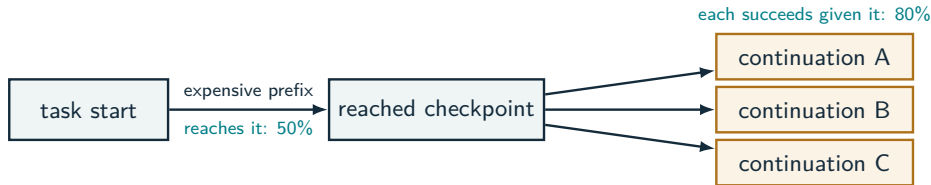


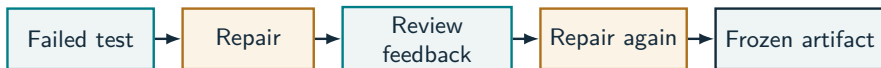
Illustration. end-to-end success = $0.5 \times 0.8 = 0.4$, while a checkpoint-started run shows 0.8.

- A checkpoint distribution differs from the task-start distribution.
- When every repair inherits one plan, continuation success alone does not validate every plan choice.

Continuation success alone does not estimate how often the checkpoint is reached.

Ecoffet et al., [First return, then explore, Nature \(2021\)](#). Illustrative probabilities.

Repair tests and generalization tests support different claims.



- A regression check supports the deterministic behavior it asserts.
- Feedback used in development changes the candidate.
- Generalization and selection claims need an appropriate untouched evaluation.

Dwork et al., Science 2015; Russo & Zou, AISTATS 2016; Kimi K3 §4.2.6; factory records.

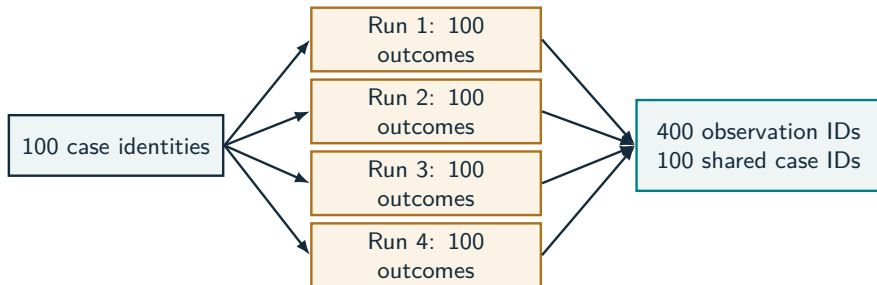
A check can run correctly and still miss the required behavior.

Question	Same-owner adoption case	Mechanism
Execution integrity	Did the check run on this digest?	Artifact-bound recorder
Behavioral coverage	Did it force CellBusy adoption and assert one dispatch?	Deterministic targeted test
Statistical support	What broader claim do repeated outcomes support?	Suitable sampling and dependence model

Duplicate rejection cannot detect a missing behavioral assertion.

SELFHOST-2 review records; proposed acceptance contract.

Four runs of 100 cases produce 400 outcomes in 100 case clusters.



Assign a new observation ID to a fresh outcome and retain its case and ancestry.

Illustrative evaluation ledger; replicate identities in Gilad et al., Communications Biology (2021).

At correlation 0.1, 100 measurements have the mean precision of about nine.

$$\text{Var}(\bar{Y}) = \sigma^2 \left[\rho + \frac{1-\rho}{N} \right]$$

$$N_{\text{eff}} = \frac{N}{1+(N-1)\rho} = \frac{100}{1+99(0.1)} \approx 9.17$$

- Equal marginal variances; common pairwise correlation.
- Describes variance-equivalent precision of a mean.
- Does not remove shared bias or give a probability of correctness.

Variance calculation; Kish, Survey Sampling, 1965. Values are illustrative.

Zero pairwise correlation does not fix the all-wrong probability.

Calculation. Three candidates. In each vector, 1 = candidate wrong. Listed vectors are equally likely.

Error vectors	Individual error rate	Pairwise correlation	All wrong
{000, 011, 101, 110}	50%	0	0%
all eight vectors	50%	0	12.5%
{001, 010, 100, 111}	50%	0	25%

Same marginals and zero pairwise correlation, yet the all-wrong rate ranges from 0% to 25%.

A selector that returns one candidate's answer succeeds with probability at most $1 - P(\text{all wrong})$.

An effective sample size describes the precision of an average. It is not a guarantee for search.

Search coverage depends on joint failure; selection is an additional task.

Calculation by enumeration. Ceiling: [Chen, arXiv:2606.27288 \(2026 preprint\)](#), ceiling $1 - \beta$, β the all-fail rate.

Shared conditions can reduce uncertainty in a paired comparison.

Compare A and B on the same case and controlled conditions.

$$\text{Var}(A - B) = \text{Var}(A) + \text{Var}(B) - 2 \text{Cov}(A, B)$$

- Positive shared variation can reduce variance of the difference.
- Record ancestry separately from shared fixtures, tests and feedback.
- Control the relevant randomness; guest seeds do not control a hosted model.

The useful sharing policy depends on the quantity being estimated.

Common-random-numbers methodology; covariance identity; proposed runtime manifest.

Passing every pair does not establish that the full composition passes.

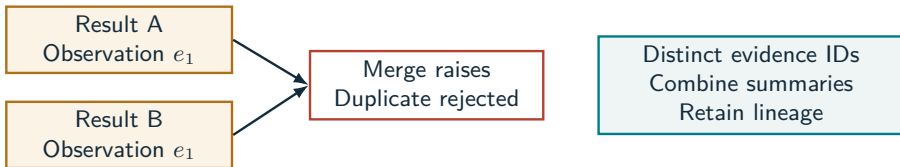
Illustration: each patch enables one worker; capacity permits two.

Patches applied	Workers	Capacity check
Each single patch	1	✓ pass
Every pair	2	✓ pass
All three	3	✗ fail

- Eight candidates have 28 pairs and 256 subsets.
- Pairwise conflict checks can miss higher-order effects.
- Verify the exact selected composition and bind its digest.

Illustrative capacity example; composition requirement in proposed branch contract.

The reducer rejects a repeated evidence ID and carries lineage forward.



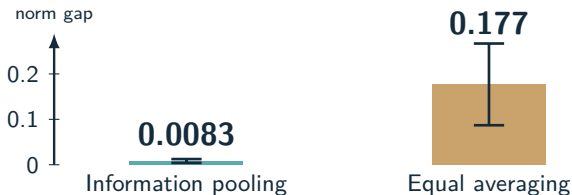
Result record: estimate, information, sample count, evidence IDs, lineage and execution metadata.

Evidence identity makes duplicate rejection an enforceable operation.

Reference reducer; Evidence-Aware Reduction for Forkable Compute.

Information pooling reduced the synthetic MLE gap from 0.177 to 0.0083.

Five logistic-regression shards: $n = 60, 120, 400, 2,000, 5,000$. Lower distance to the full-data MLE is better.



Eight fixed seeds; whiskers are SD. Common parameter and calibrated information assumed.

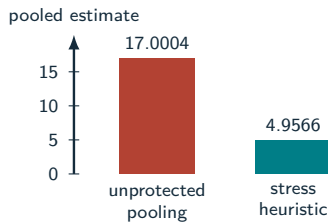
Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689) (2026), §4: synthetic checks.

Inflated reported precision pushed the synthetic pooled estimate to 17.0004.

One shard inflates its precision

A 2,000-point shard returns a distant estimate and inflates its reported precision a further 50×.

- Claimed precision raises the shard's weight.
- Sample count alone cannot detect inflated information per point.
- The stress heuristic is not a Byzantine guarantee.



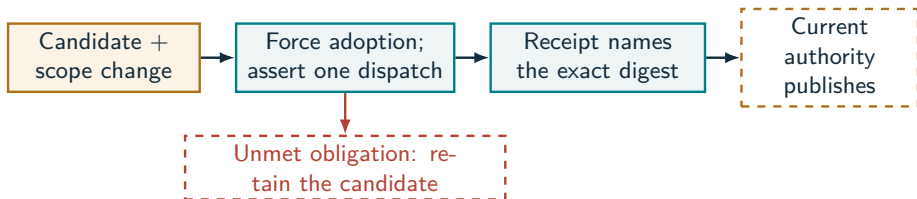
DESIGN The controller sets each shard's precision by trusted recomputation or calibrated external scoring.

Evidence identity and justified statistical weight are separate requirements.

Synthetic check, Eliaz, [arXiv:2607.09689](https://arxiv.org/abs/2607.09689) (2026), §4. Synthetic numerical stress case.

The publication gate requires the adoption test on the exact candidate.

The new same-owner adoption path is a concrete acceptance obligation.



Selection or composition must precede verification of the artifact that will be published.

Export the patch and receipts on every exit, including refusal.

Gate design motivated by SELFHOST-2 and SELFHOST-3.

20 batches at each fan-out compare restore with a warm cached template.

	Warm template	Restore
Preparation	Dependencies baked in; pinned source	Capture after setup and one warm test
Fan-out	$N \in \{3, 6, 12\}$; 20 interleaved batches per arm and N	
Endpoint	Batch request $\rightarrow$ last child's first test result	

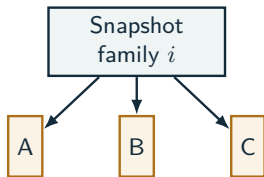
- No model calls; cold start is a reference.
- Establish declared restore fidelity before timing.
- Report capture cost, failed restores and batch latency separately.

A timing win supports this restore path on this workload.

Experiment design: H1 in PREREGISTRATION.md; matched state and the last-child endpoint.

12 snapshot families compare sibling and separately prepared co-failures.

Shared parent



Separately prepared parents

Three restores from families $i + 1$, $i + 2$, $i + 3$; matched host and start slot.

12 families; at least 300 lockstep rounds.

Record binary outcome and failure class.

Identify a parent-state feature that capture preserves and the race test consumes.

Experiment design: H2 in PREREGISTRATION.md; environmental failure outcomes.

At correlation 0.10, nine outcomes have a variance-equivalent count of five.

- Reusing snapshot families links comparisons.
- Repeated rounds can share scheduling and temporal effects.
- Declare independent blocks or a justified dependence model before collection.

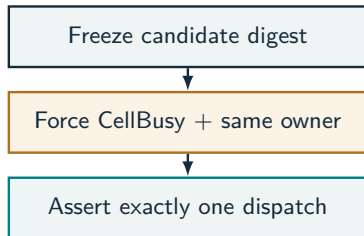
Nine equal-variance measurements	Absolute ρ	N_{eff}
Separately prepared restores	0.10	5.00
Shared-parent restores	0.15	4.09

An excess $\Delta\rho = 0.05$ is insufficient without the baseline dependence.

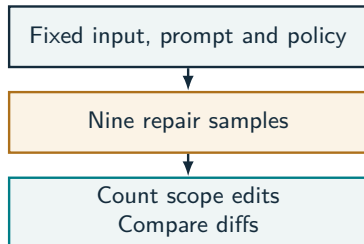
Illustrative equal-variance calculation; dependence analysis is a proposed protocol amendment.

One adoption test checks dispatch; nine repair samples measure repeatability.

Behavioral obligation



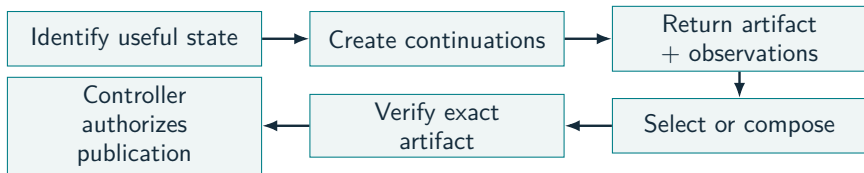
Repair repeatability



The acceptance test checks the artifact; resampling describes the repair process.

Experiment design: regression obligation and Phase C in PREREGISTRATION.md.

Reuse the useful state, retain the evidence, and verify the exact result.



State: AFL++
10–20× reported speedup

Evidence: information pooling
MLE gap 0.177 → 0.0083

Verification: CyberGym
56 crashes → 22 zero-days

AFL++ documentation; Eliaz, arXiv:2607.09689; CyberGym report. Integrated lifecycle is the design developed in this talk.